# PROVINCIAL EXAMINATION

# NOVEMBER 2023

# GRADE 6

## NATURAL SCIENCES AND TECHNOLOGY



**TIME: 1½ hours**

**MARKS: 60**

**12 pages**

| | | | | |
|---|---|---|---|---|
| **DISTRICT** | | | | |
| **SCHOOL NAME** | | | | |
| **EMIS NUMBER** | | | | |
| **CLASS (e.g. 6A)** | | | | |
| **NAME AND SURNAME** | | | | |
| **GENDER** | BOY | | GIRL | |
| **DATE** | | | | |

| **SECTION** | **A** | | | | | | **B** | | | **C** | | **TOTAL** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Question** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | |
| **Mark allocation** | **5** | **5** | **5** | **5** | **5** | **5** | **5** | **7** | **9** | **3** | **6** | **60** |
| **Learner's Marks** | | | | | | | | | | | | |

## INSTRUCTIONS AND INFORMATION

**This question paper consists of THREE Sections, A, B and C.**

1. Answer ALL three sections.
2. ALL questions must be answered on the question paper in the spaces provided.
3. Write neatly and legibly.

**STRANDS**

| | |
|---|---|
| NATURAL SCIENCE | Energy and Change, Planet Earth & Beyond |
| TECHNOLOGY | Systems and control |

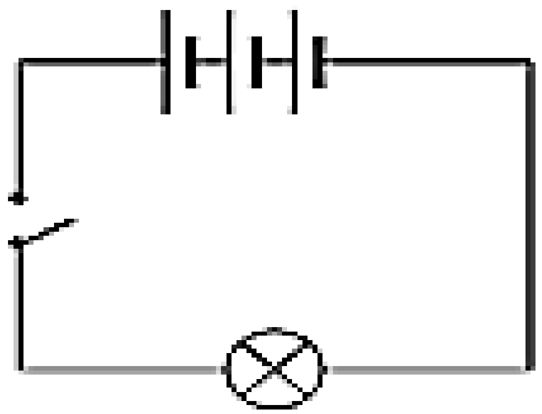
The question paper consists of **SECTION A**, **SECTION B AND SECTION C.**

| **SECTION A:** LOW ORDER QUESTIONS/COGNITIVE LEVEL 1 | **SECTION B:** MIDDLE ORDER QUESTIONS/ COGNITIVE LEVEL (2 & 3) | **SECTION C:** HIGH ORDER QUESTIONS/COGNITIVE LEVEL (4, 5 & 6) |
|---|---|---|
| **Q1:** Electric circuits | **Q7:** How coal is formed? | **Q10:** Systems to explore the Moon and Mars |
| **Q2:** Mains electricity; Conductors and Insulators; Renewable ways to generate electricity; Electric circuits | **Q8:** Mains electricity, cost of electricity, using electricity safely | **Q11:** Telescopes |
| **Q3:** Telescopes, Systems to explore the Moon and Mars | **Q9:** Electric circuits; Circuit diagrams; Using electric circuits | |
| **Q4:** Systems to solve problems | | |
| **Q5:** The Sun, planets and asteroids; The rotation and revolution of the Earth; Telescopes | | |
| **Q6:** Movement of the Earth and Planets | | |
| **Total – 30** | **Total – 21** | **Total – 9** |

## SECTION A

### QUESTION 1: ELECTRIC CIRCUITS

Look at the circuit diagram below and answer the following questions.



1.1 How many cells are in the circuit? ______________________________ (1)

1.2 What is the function of the battery (or cells) in the circuit?

______________________________ (1)

1.3 Does the circuit have a switch? ______________________________ (1)

1.4 Name the type of output energy in the circuit. ______________________________ (1)

1.5 Which device produces the output energy in the circuit? ______________ (1)

**[5]**

### QUESTION 2

Four possible answers are given for each question below. Read each question and circle the correct answer e.g. 2.6 A.

2.1 Which of the following is an example of a non-renewable resource used to generate electricity?

A Solar power
B Hydro-electric power
C A coal-burning power station
D Wind power

2.2 Parts of electric circuits are called …

A appliances.
B circuit diagram.
C copper wire.
D components.

2.3 The main use of coal during the generation of electricity is to …

A turn turbines.
B heat the pylons.
C heat the water.
D change the energy of the moving turbine.

2.4 Which of the following is NOT an example of fossil fuel?

A Water
B Coal
C Natural gas
D Oil

2.5 This device converts electrical energy into movement energy.

A Bulb
B Motor
C Buzzer
D Battery

**[5]**

**QUESTION 3: TELESCOPES, SYSTEMS TO EXPLORE THE MOON AND MARS**

Match the number in COLUMN A with the letter COLUMN B by writing the correct letter in the ANSWER COLUMN below.

| COLUMN A | | COLUMN B | | Answer |
|---|---|---|---|---|
| **E.g.** | **Satellite** | A | Without people aboard to operate the spacecraft | **E.g. – G** |
| 3.1 | Sojourner | B | The first robotic rover to land on another planet | 3.1 |
| 3.2 | Telescope | C | Is used to look into space and gather information | 3.2 |
| 3.3 | Unmanned | D | The new robotic rover for NASA | 3.3 |
| 3.4 | Space Rover | E | Is used to collect rock and soil samples and take photographs of the surface of the Moon and Mars | 3.4 |
| 3.5 | Curiosity | F | An object intentionally placed into orbit in outer space | 3.5 |

**[5]**

## QUESTION 4: THE SOLAR SYSTEM

Indicate whether the following statements are TRUE or FALSE.

4.1 Mars is also known as the "Red Planet". ______________ (1)

4.2 The largest planet in the solar system is Jupiter. ______________ (1)

4.3 Planets closest to the sun are called "inner planets". ______________ (1)

4.4 There are 8 planets in the solar system. ______________ (1)

4.5 The rocky space objects that orbit the sun in the zone between Mars and Jupiter are called satellites. ______________ (1)

**[5]**

## QUESTION 5

Provide the scientific name/term for each of the following statements.

| | Statement | Answer |
|---|---|---|
| 5.1 | The movement of one body around another in space | |
| 5.2 | A rocky body that travels around a planet | |
| 5.3 | The line that passes through the centre of an object | |
| 5.4 | A large circular hole with a rim around it | |
| 5.5 | An instrument that magnifies distant objects | |

**[5]**

**QUESTION 6: LABELLING STRUCTURES**

Label the parts of a light bulb.

Choose the words from the word box below to correctly label the parts of a light bulb.

| | | | | |
|---|---|---|---|---|
| Electrical contact; | Filament; | Contact wire; | Cap; | Tungsten Filament |


6.1 ________________ (1)

6.2 ________________ (1)

6.3 ________________ (1)

6.4 ________________ (1)

6.5 ________________ (1)

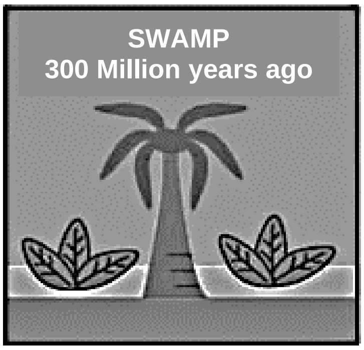
**[5]**

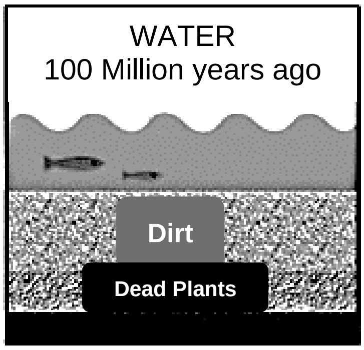
**TOTAL SECTION A: 30**

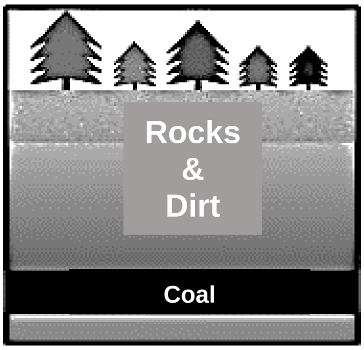
## SECTION B

## QUESTION 7: FOSSILS

The pictures below show how coal is formed.




Rearrange the steps below to show the correct order in which coal is formed. Write only the LETTER to show the correct order.

| | Incorrect order | Correct order |
|---|---|---|
| A | Over millions of years decomposed plants fossilise to become coal. | |
| B | Plants use light energy from the sun to grow. | |
| C | The sun gives heat and light energy. | |
| D | The plants die and decompose. | |
| E | Layers of sand and mud are deposited on top of decomposed plants over millions of years. | |

**[5]**

## QUESTION 8: MAINS ELECTRICITY

8.1 Cost of electricity.

Kagiso compiled a table to show the amount of electricity used by some appliances in his home.

| Appliance | Electricity use by an appliance (W) |
|---|---|
| Electric Stove | 8 000 |
| Geyser | 4 000 |
| Television | 340 |
| Kettle | 2 000 |
| Fridge | 150 |

(a) Explain why the electric stove and the geyser use more electrical energy than the television and the fridge.

______________________________________________

______________________________________________ (2)

(b) Suggest TWO ways in which Kagiso's family could save electricity.

1 ______________________________________________

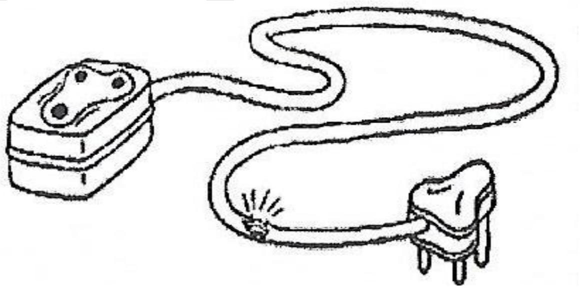
2 ______________________________________________ (2)

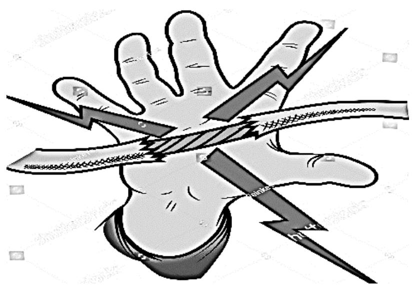
(c) Provide an example of a renewable source of electricity that Kagiso could tell his family to invest in.

______________________________________________ (1)

8.2 Using electricity safely.

Look at the pictures below and answer the following questions.





(a) Explain why cables such as the ones above cannot be used.

______________________________________________ (1)

(b) If you were an electrician, what safety measure would you put in place if you were working with an exposed wire?

_______________________________________________ (1)
**[7]**

**QUESTION 9: ELECTRIC CIRCUITS**

Study the picture of the real circuit below and answer the questions that follow.

9.1 Illustrate and label a circuit diagram to represent the above circuit in the space below.

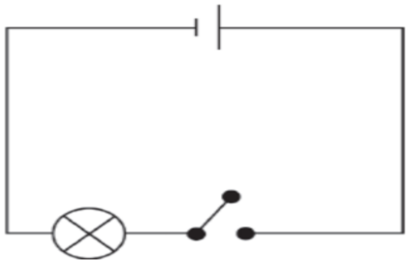
(4)

9.2 Illustrate a systems diagram for the above electrical system in the spaces below.

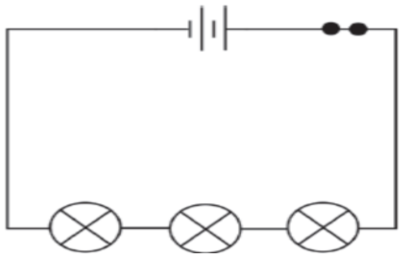
| **INPUT** | → | **PROCESS** | → | **OUTPUT** |
|---|---|---|---|---|
| ________________ | | Electrical charges move through the circuit | | ________________ |

(2)

9.3 Compare the circuit diagrams below and answer the following question.



**Circuit A**



**Circuit B**

Explain which circuit will provide an output. Give a reason for your answer.

___________________________________________________________________

___________________________________________________________________

___________________________________________________________________ (3)

**[9]**

**TOTAL SECTION B: 21**

## SECTION C

### QUESTION 10: SYSTEMS TO EXPLORE THE MOON AND MARS

10.1 Briefly discuss TWO ways in which the Lunar Roving Vehicle (LRV) assisted astronauts with their scientific discoveries.

1 ____________________________________________

____________________________________________

2 ____________________________________________

____________________________________________ (2)

10.2 Conclude why LRV's are sent to Mars instead of astronauts.

____________________________________________ (1)
**[3]**

### QUESTION 11: TELESCOPES

Read the following passage and answer the questions that follow.

***The Square Kilometer Array (SKA)***

South Africa has spent eight years preparing to host the world's largest telescope. The other country wanting to host the SKA is Australia.

We have a very good site in South Africa. The SKA radio telescope will have several antennae and dishes placed in various areas. This will send information to a central computer to produce a picture.

The main centre will be in the Karoo with other centres away from people and away from FM broadcasting and cellphone towers to prevent interferences.

The SKA will be 50 times more sensitive and be able to survey the sky 10 000 times faster than any other telescope.

The telescope will observe black holes, stars and galaxies or bands of stars.
This will be good for the economy and for job creation.

11.1 Outline what factors scientists would consider when identifying a main centre for building a telescope.

____________________________________________

____________________________________________ (2)

11.2 Suggest TWO reasons why the SKA will be better than other telescopes.

1 ______________________________________________

______________________________________________

2 ______________________________________________

______________________________________________ (2)

11.3 You are required to make a telescope for the NST Science Fair. Your teacher has given you a kitchen towel roll and the bottom of a glass bottle. Discuss how the bottom of the glass bottle aids in making a telescope.

______________________________________________

______________________________________________

______________________________________________

______________________________________________ (2)

**[6]**

**TOTAL SECTION C: 9**

**TOTAL: 60**



**END**